

Detector Acceptance Study

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- Files Used & Event Selection
- Invariant Mass Studies
- x_F Studies (Yields & Acceptances)
- p_T Studies (Yields & Acceptances)
- Binned Slice Studies (x_F and p_T bins)

Files Used & Kinematic Bins

Files Used:

- LH2 and LD2 Drell-Yan Monte Carlo (newly made by Kenichi)
- **Thrown:** `rs67_<lh2/ld2>.acc.root:tree_4pi:event`
- **Accepted:** `rs67_<lh2/ld2>.acc.root:tree_acc:event`

Acceptance Calculation:

$$\text{Acceptance} = \frac{\sum w_{\text{accepted}}}{\sum w_{\text{thrown}}} \quad \text{where } w = \text{weight}$$

Kinematic Bin Boundaries:

- **Mass [GeV]:** [4.2, 4.5, 4.8, 5.1, 5.4, 5.7, 6.0, 6.3, 6.6, 6.9, 7.5, 8.8]
- x_F : [-0.05, 0.0, 0.05, ..., 0.85]
- p_T [GeV/c]: [0.0, 0.32, 0.49, 0.63, 0.77, 0.95, 1.18, 1.8]

Event Selection Criteria (Chuck's Cuts)

Dimuon Kinematics:

- $4.2 < \text{Mass} < 8.8 \text{ GeV}$
- $-0.1 < x_F < 0.95$
- $0.05 < x_T \leq 0.58$
- $38 < dp_z < 116 \text{ GeV}$
- $dp_x^2 + dp_y^2 < 5.0 \text{ GeV}^2$

Target & Vertex:

- $-280 < dz < -5 \text{ cm}$
- $|dx| < 0.25 \text{ cm}$
- $|dy - 1.6| < 0.22 \text{ cm}$

Track Quality:

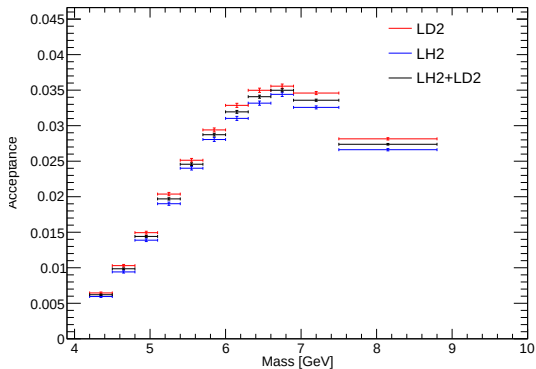
- $\chi_{\text{dimuon}}^2 < 18$
- Track separation $< 270 \text{ cm}$
- $p_{z,\text{st1}} \in [9, 75] \text{ GeV}$
- $n\text{Hits}_1 > 13, n\text{Hits}_2 > 13$

Generator-Level Fiducial Cuts (Thrown):

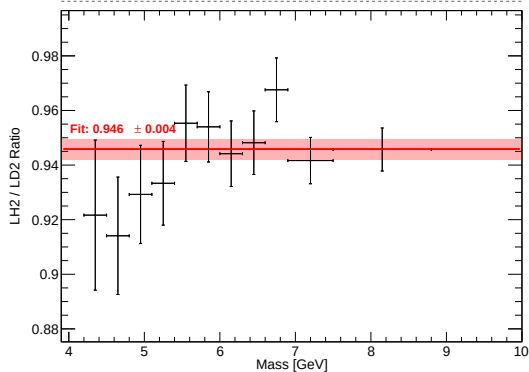
- $4.2 < \text{Mass} < 8.8 \text{ GeV}$
- $-0.1 < x_F < 0.95$
- $0.0 < p_T \leq 3.0 \text{ GeV}/c$

Overlay & Ratio: Mass (Integrated)

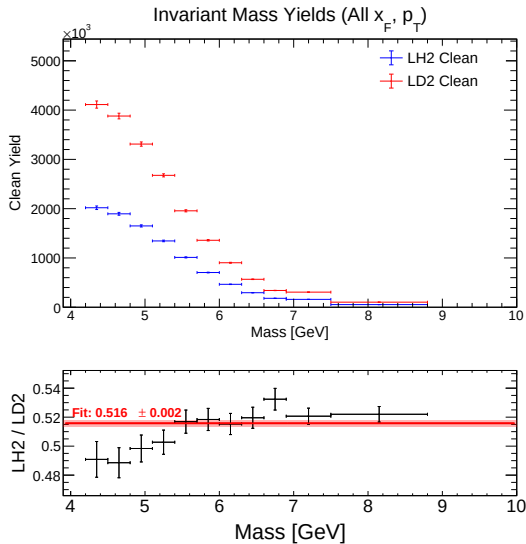
Integrated Acceptance vs mass



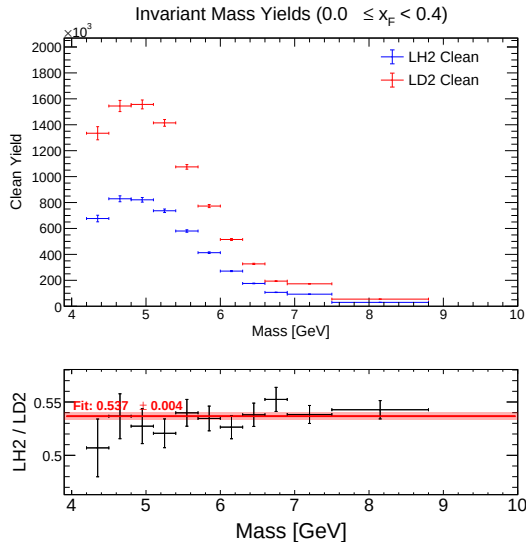
LH2/LD2 Integrated Acceptance vs mass



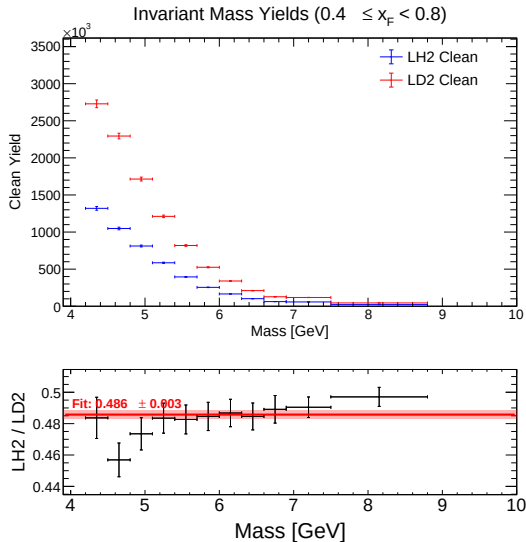
Invariant Mass Yields & Ratio (All x_F , p_T)



Invariant Mass Yields & Ratio ($0.0 < x_F \leq 0.4$)

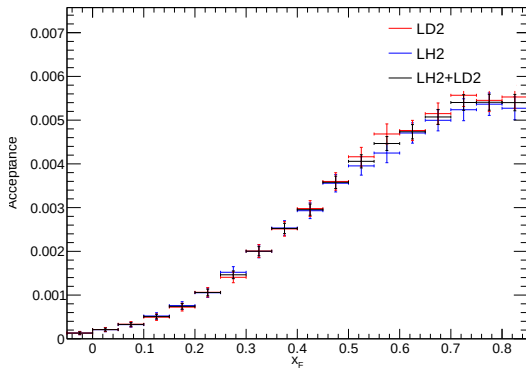


Invariant Mass Yields & Ratio ($0.4 < x_F \leq 0.8$)

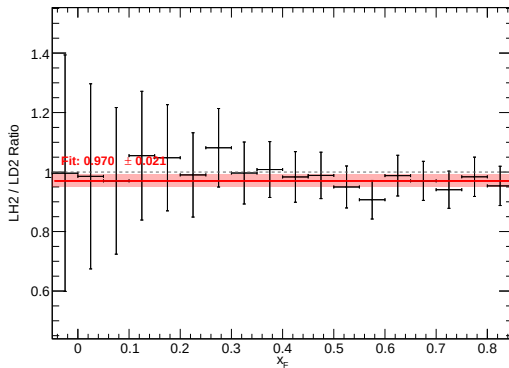


Overlay & Ratio: x_F (Integrated)

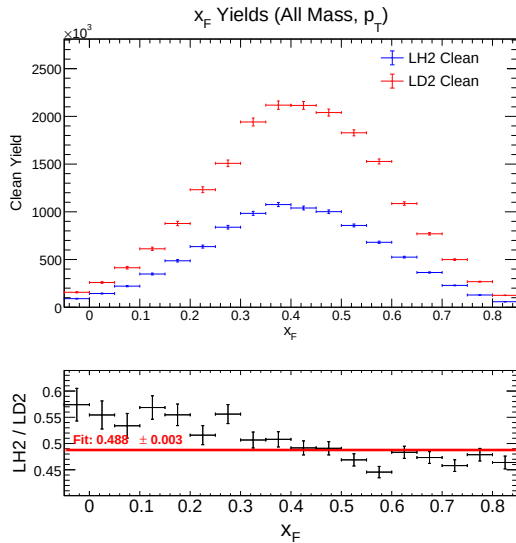
Integrated Acceptance vs x_F



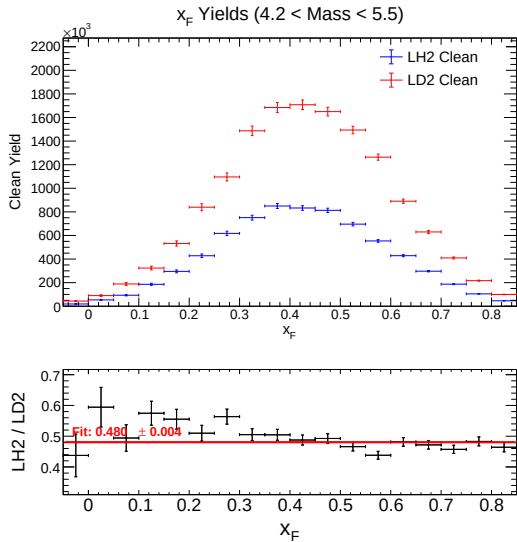
LH2/LD2 Integrated Acceptance vs x_F



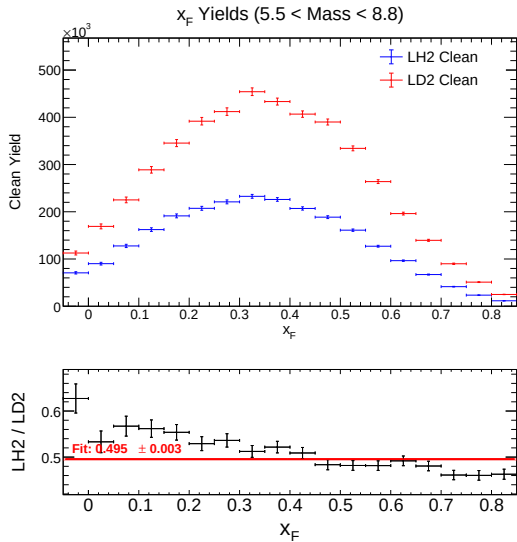
x_F Yields & Ratio (All Mass, p_T)



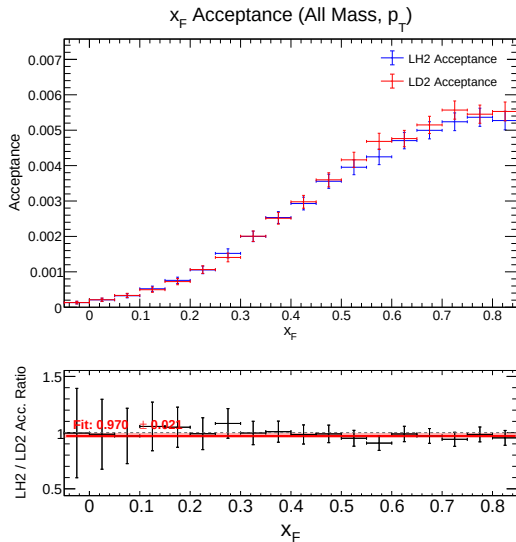
x_F Yields & Ratio ($4.2 < \text{Mass} \leq 5.5$)



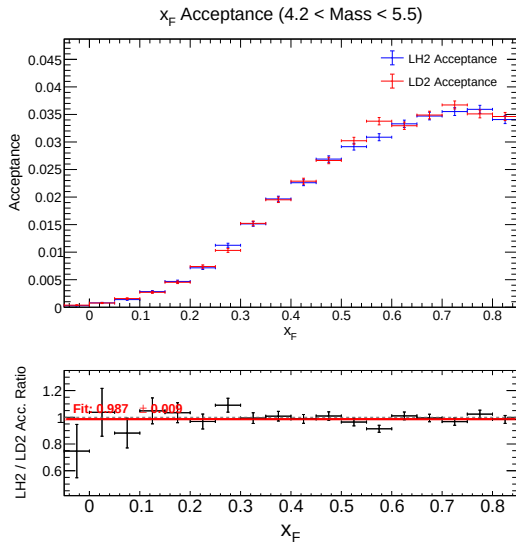
x_F Yields & Ratio ($5.5 < \text{Mass} \leq 8.8$)



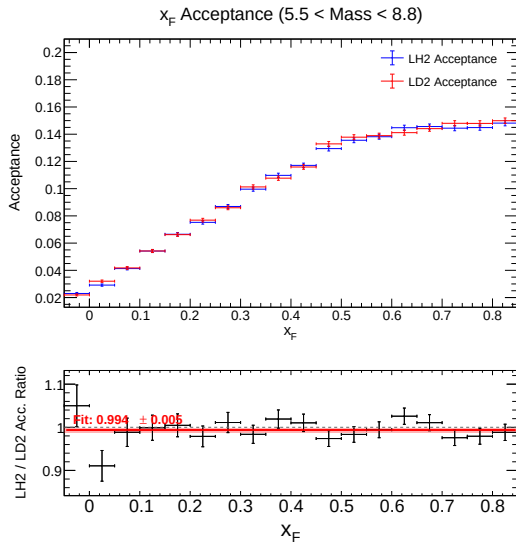
x_F Acceptance & Ratio (All Mass, p_T)



x_F Acceptance & Ratio ($4.2 < \text{Mass} < 5.5$)

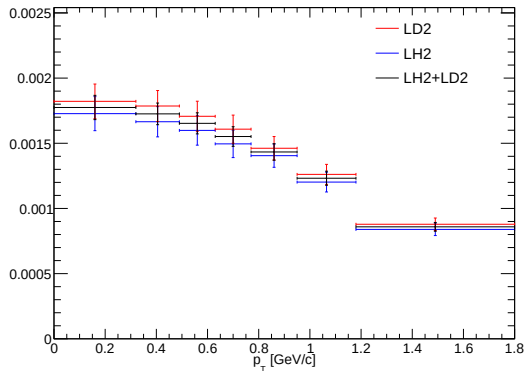


x_F Acceptance & Ratio ($5.5 < \text{Mass} < 8.8$)

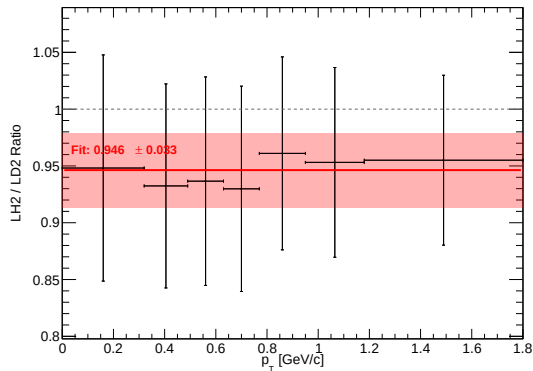


Overlay & Ratio: p_T (Integrated)

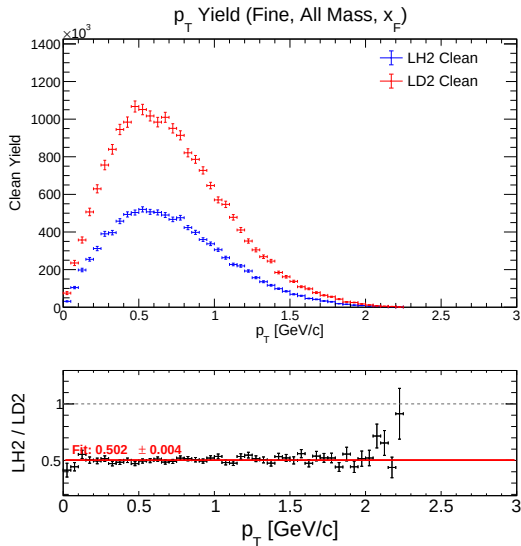
Integrated Acceptance vs p_T



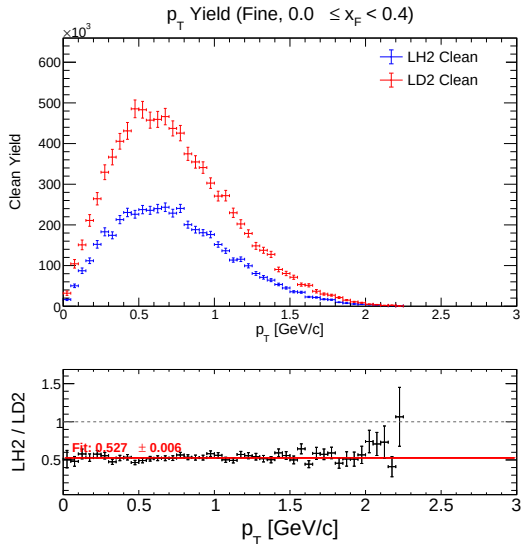
LH2/LD2 Integrated Acceptance vs p_T



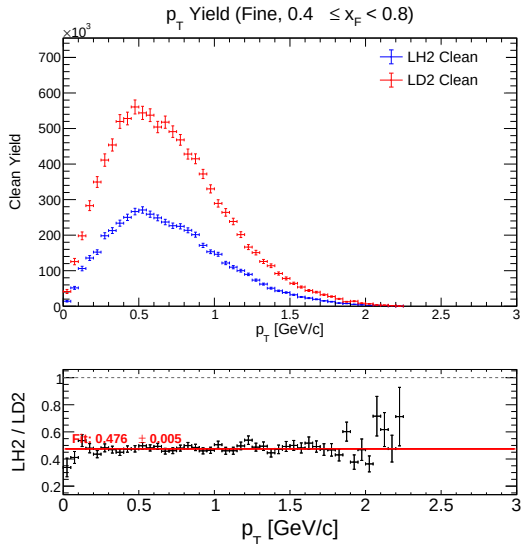
p_T Yields & Ratio (Fine Bins, All Mass, x_F)



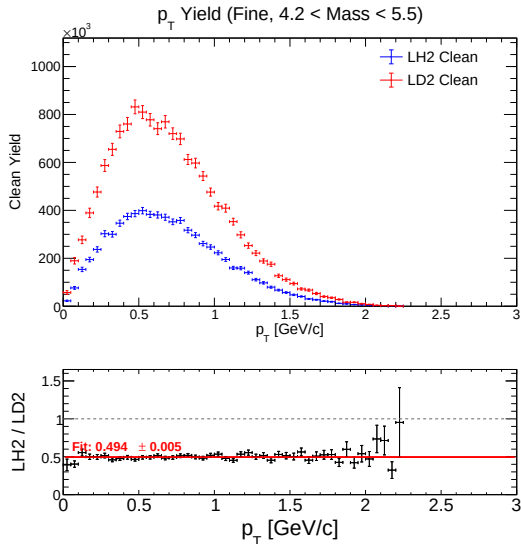
p_T Yields & Ratio (Fine Bins, $0.0 < x_F < 0.4$)



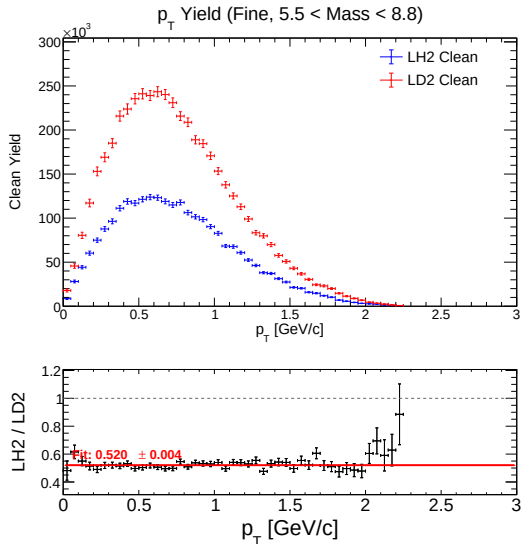
p_T Yields & Ratio (Fine Bins, $0.4 < x_F < 0.8$)



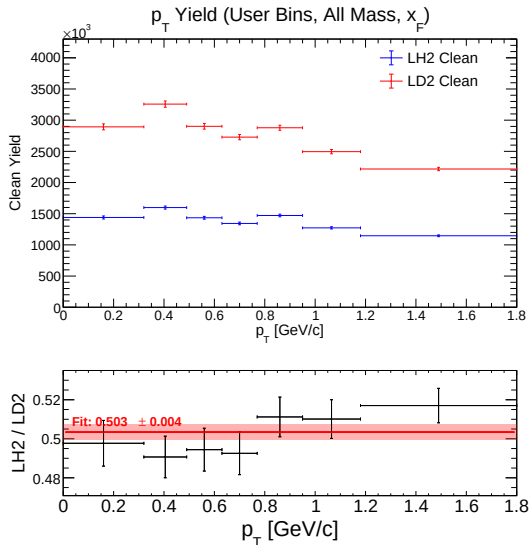
p_T Yields & Ratio (Fine Bins, $4.2 < \text{Mass} < 5.5$)



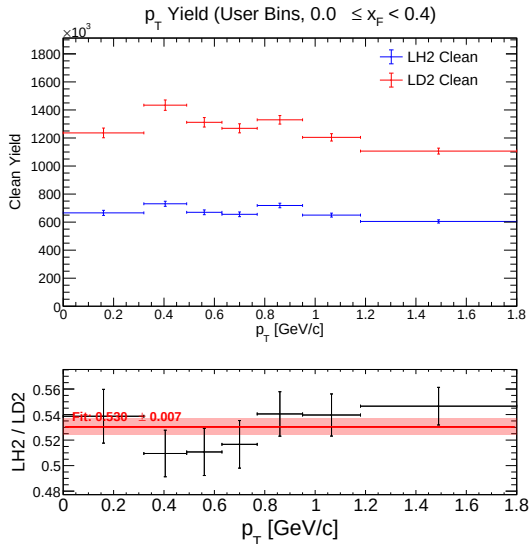
p_T Yields & Ratio (Fine Bins, $5.5 < \text{Mass} < 8.8$)



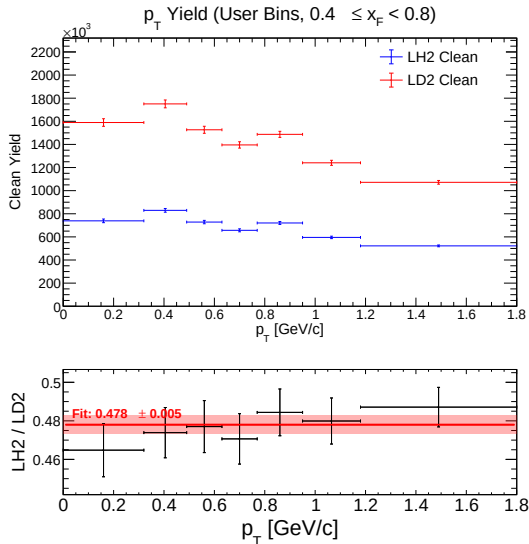
p_T Yields & Ratio (User Bins, All Mass, x_F)



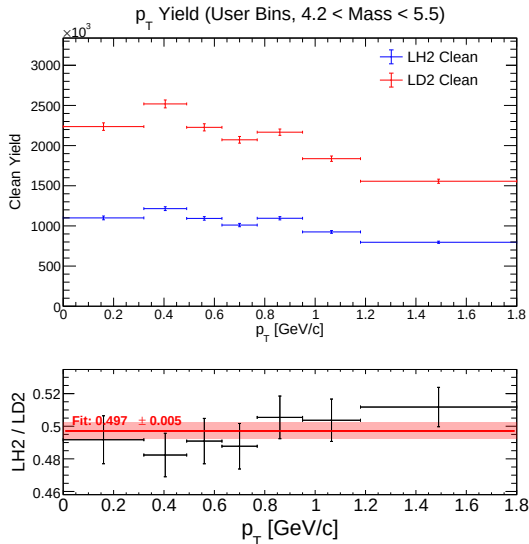
p_T Yields & Ratio (User Bins, $0.0 < x_F < 0.4$)



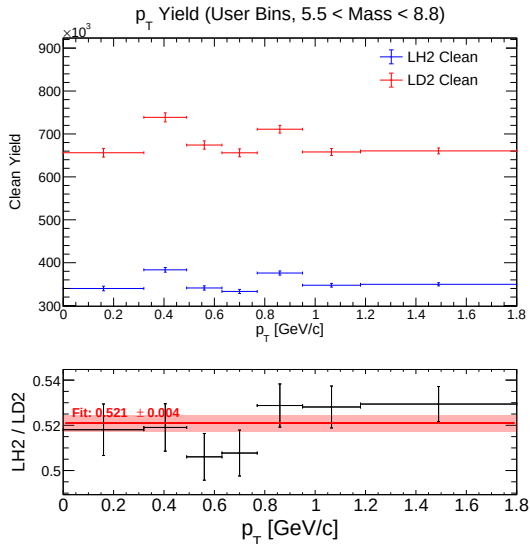
p_T Yields & Ratio (User Bins, $0.4 < x_F < 0.8$)



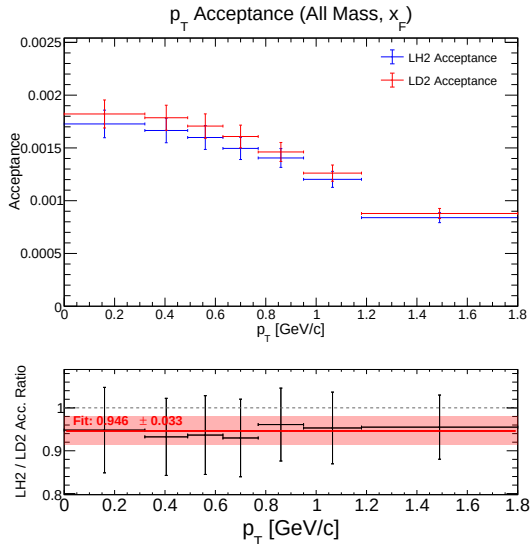
p_T Yields & Ratio (User Bins, $4.2 < \text{Mass} < 5.5$)



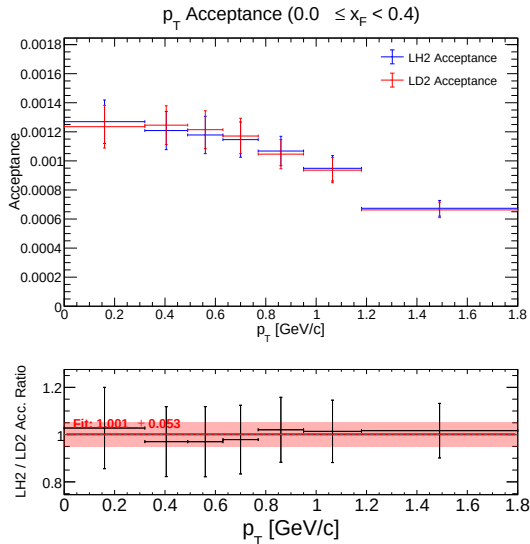
p_T Yields & Ratio (User Bins, $5.5 < \text{Mass} < 8.8$)



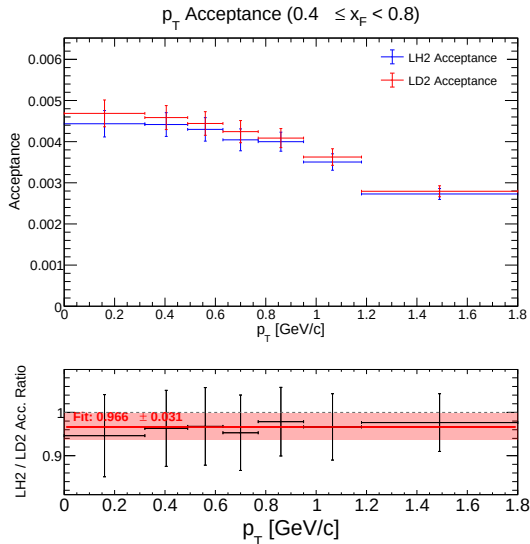
p_T Acceptance & Ratio (User Bins, All Mass, x_F)



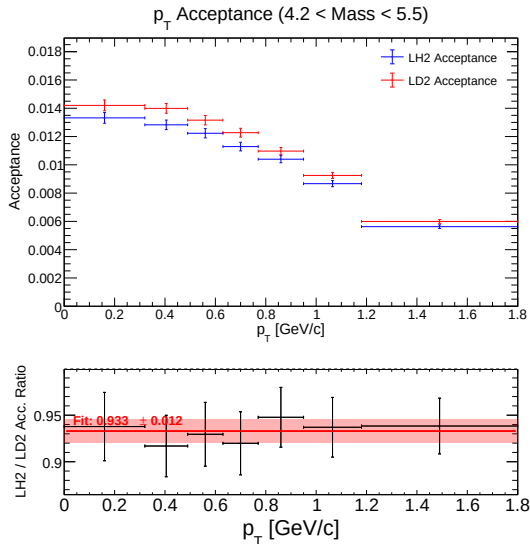
p_T Acceptance & Ratio (User Bins, $0.0 < x_F < 0.4$)



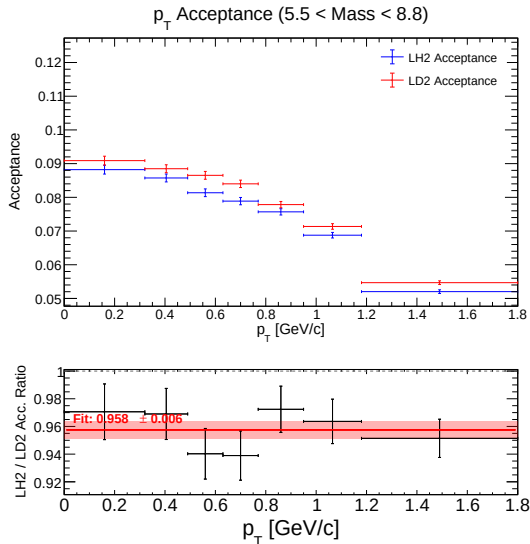
p_T Acceptance & Ratio (User Bins, $0.4 < x_F < 0.8$)



p_T Acceptance & Ratio (User Bins, $4.2 < \text{Mass} < 5.5$)

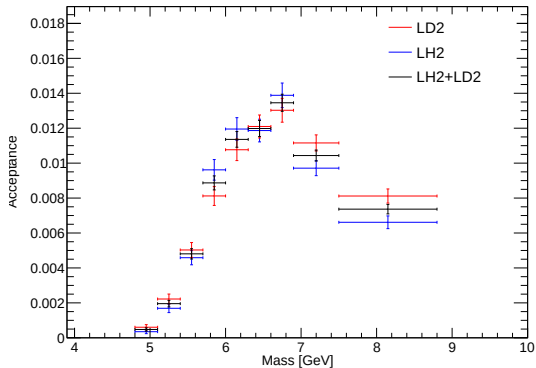


p_T Acceptance & Ratio (User Bins, $5.5 < \text{Mass} < 8.8$)

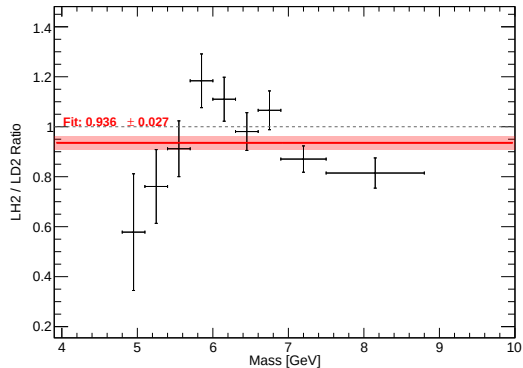


Overlay & Ratio: x_F Bin 0

$-0.05 \leq x_F < 0.00$

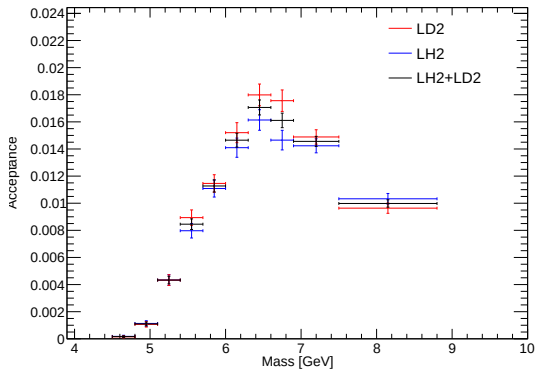


LH2/LD2 $-0.05 \leq x_F < 0.00$

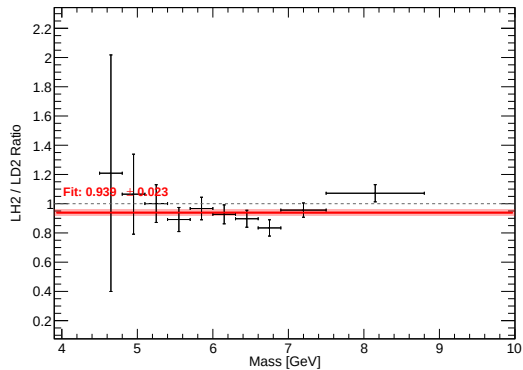


Overlay & Ratio: x_F Bin 1

$0.00 \leq x_F < 0.05$

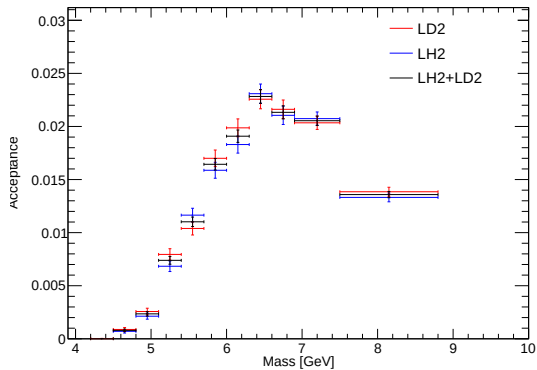


LH2/LD2 $0.00 \leq x_F < 0.05$

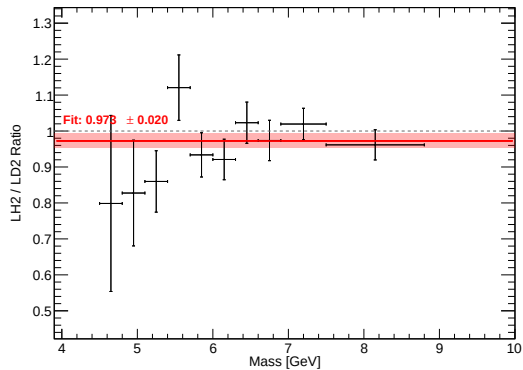


Overlay & Ratio: x_F Bin 2

$0.05 \leq x_F < 0.10$

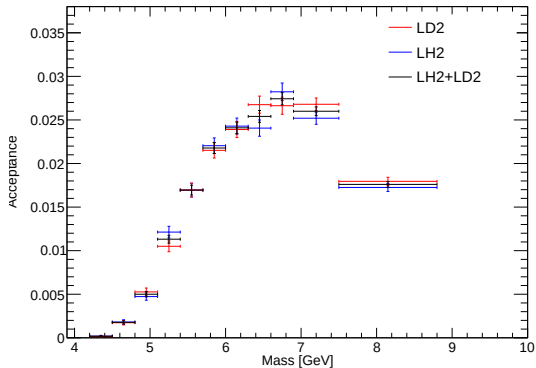


LH2/LD2 $0.05 \leq x_F < 0.10$

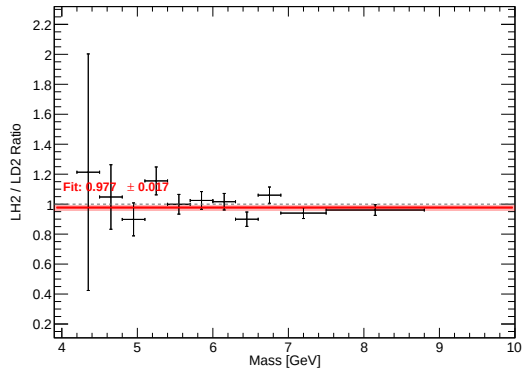


Overlay & Ratio: x_F Bin 3

$0.10 \leq x_F < 0.15$

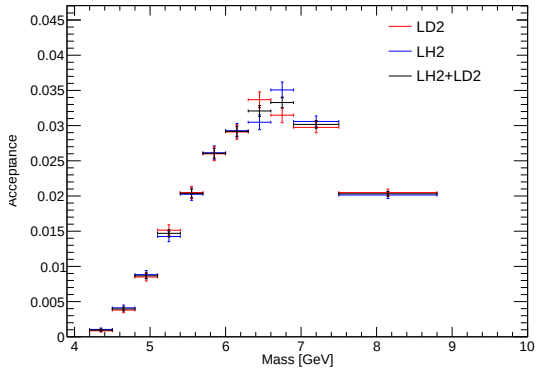


LH2/LD2 $0.10 \leq x_F < 0.15$

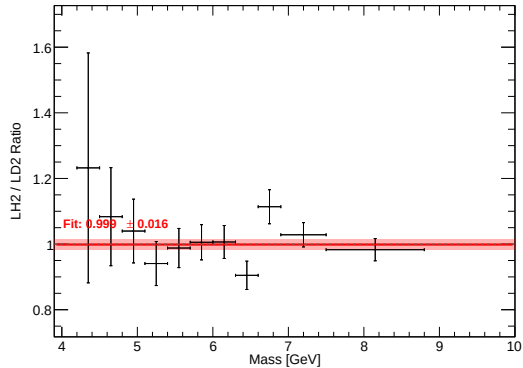


Overlay & Ratio: x_F Bin 4

$0.15 \leq x_F < 0.20$

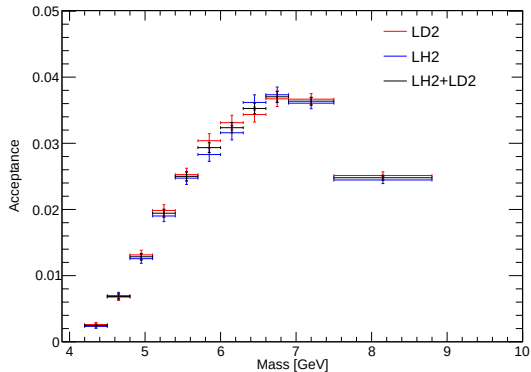


LH2/LD2 $0.15 \leq x_F < 0.20$

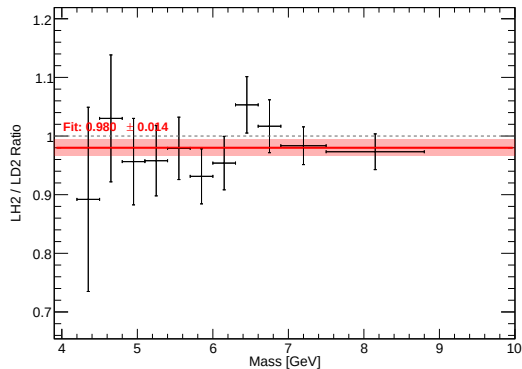


Overlay & Ratio: x_F Bin 5

$0.20 \leq x_F < 0.25$

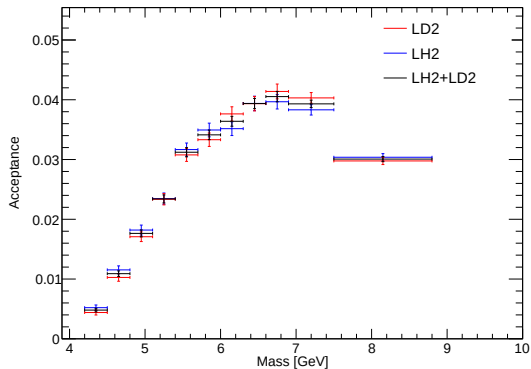


LH2/LD2 $0.20 \leq x_F < 0.25$

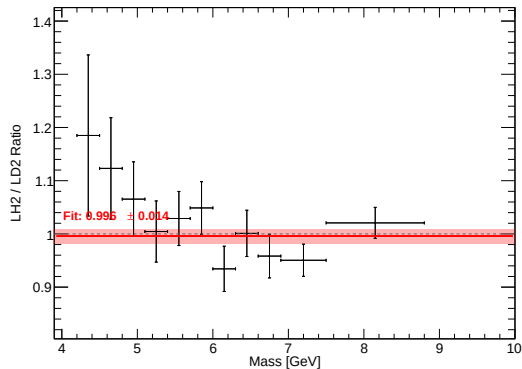


Overlay & Ratio: x_F Bin 6

$0.25 \leq x_F < 0.30$

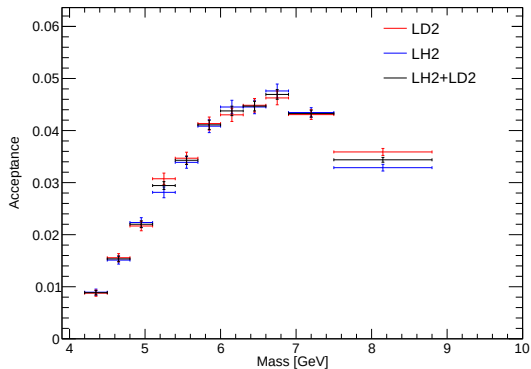


LH2/LD2 $0.25 \leq x_F < 0.30$

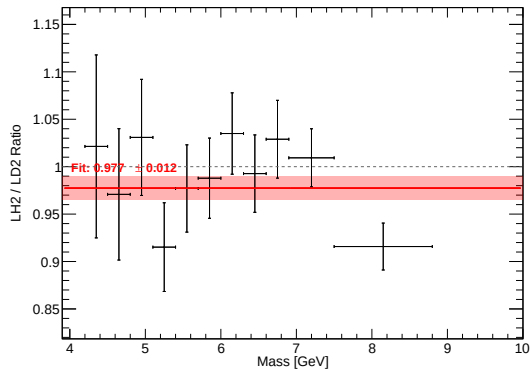


Overlay & Ratio: x_F Bin 7

$0.30 \leq x_F < 0.35$

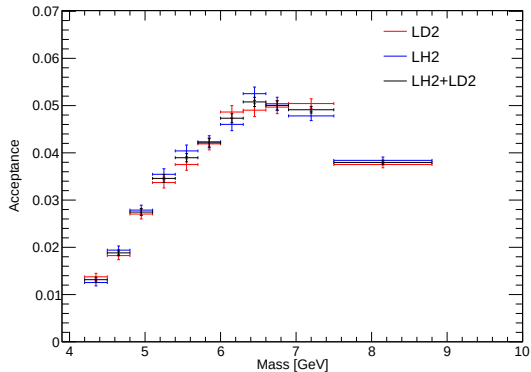


LH2/LD2 $0.30 \leq x_F < 0.35$

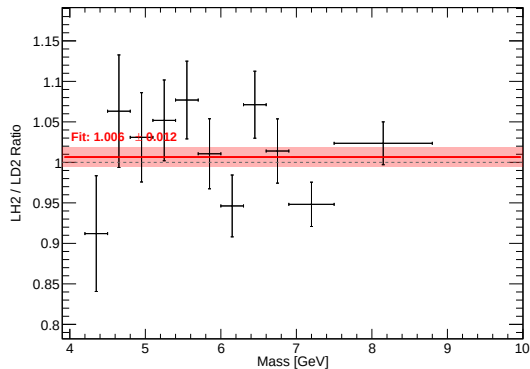


Overlay & Ratio: x_F Bin 8

$0.35 \leq x_F < 0.40$

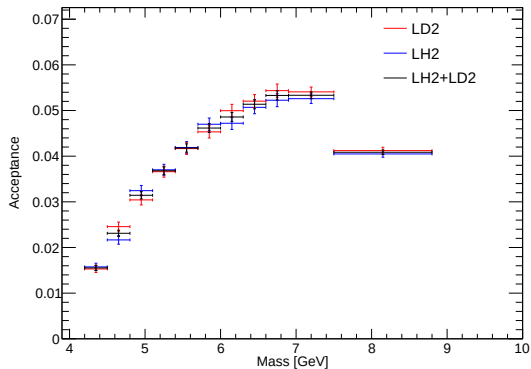


LH2/LD2 $0.35 \leq x_F < 0.40$

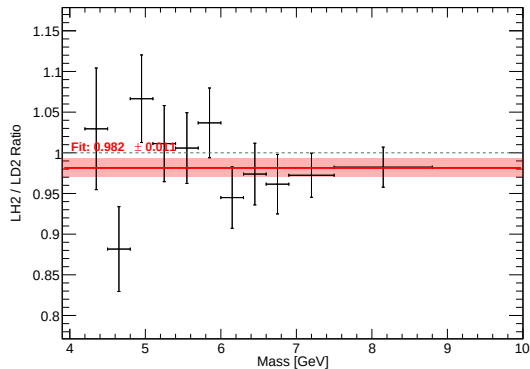


Overlay & Ratio: x_F Bin 9

$0.40 \leq x_F < 0.45$

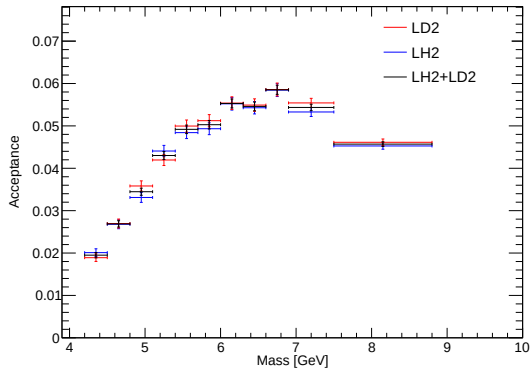


LH2/LD2 $0.40 \leq x_F < 0.45$

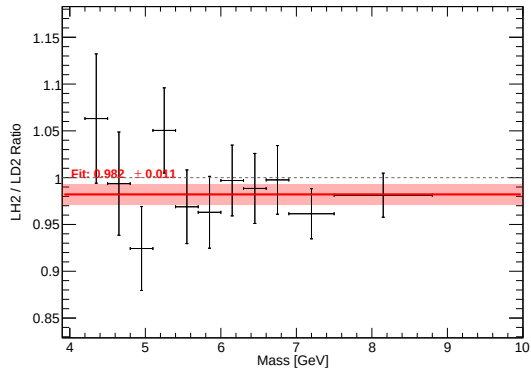


Overlay & Ratio: x_F Bin 10

$0.45 \leq x_F < 0.50$

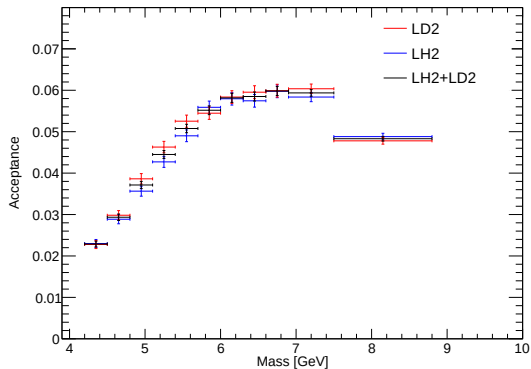


LH2/LD2 $0.45 \leq x_F < 0.50$

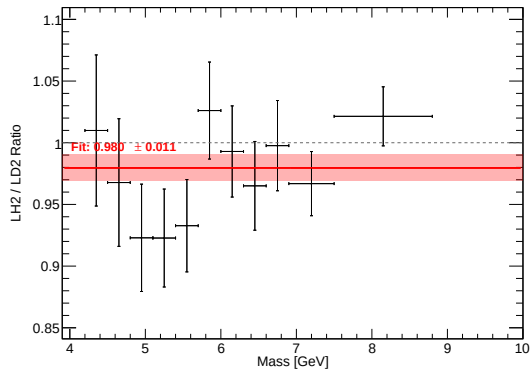


Overlay & Ratio: x_F Bin 11

$0.50 \leq x_F < 0.55$

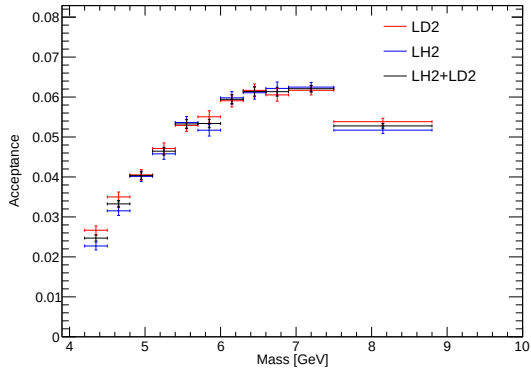


LH2/LD2 $0.50 \leq x_F < 0.55$

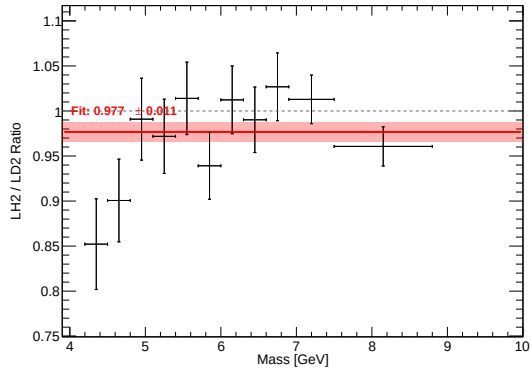


Overlay & Ratio: x_F Bin 12

$0.55 \leq x_F < 0.60$

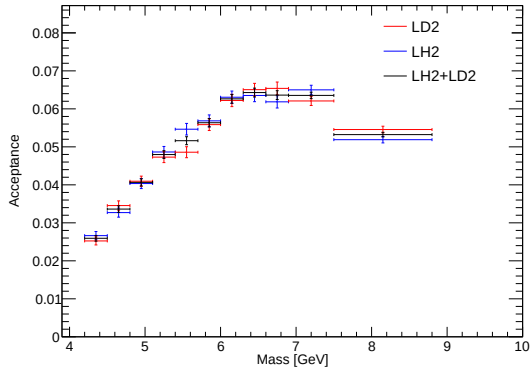


LH2/LD2 $0.55 \leq x_F < 0.60$

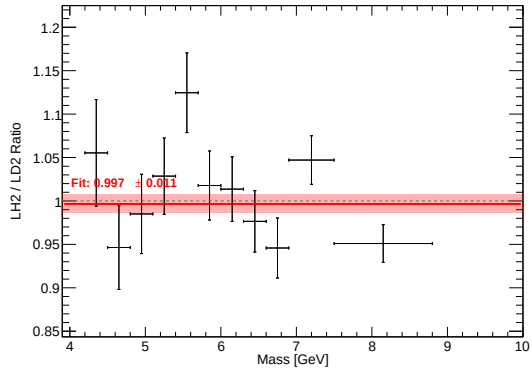


Overlay & Ratio: x_F Bin 13

$0.60 \leq x_F < 0.65$

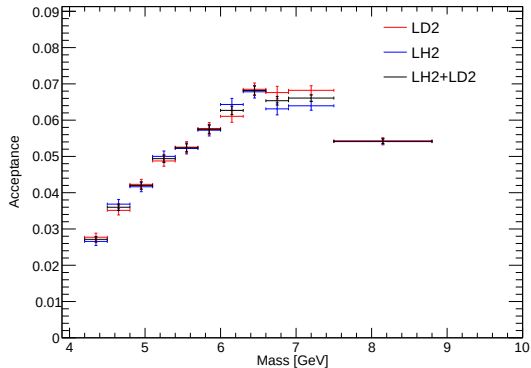


LH2/LD2 $0.60 \leq x_F < 0.65$

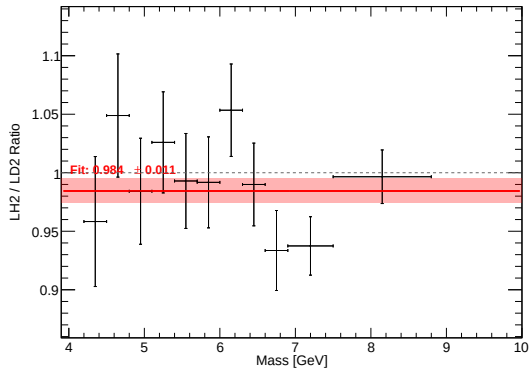


Overlay & Ratio: x_F Bin 14

$0.65 \leq x_F < 0.70$

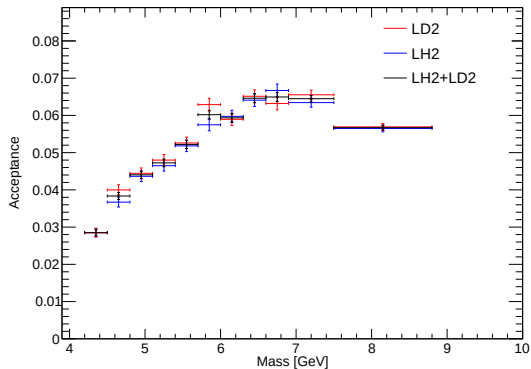


LH2/LD2 $0.65 \leq x_F < 0.70$

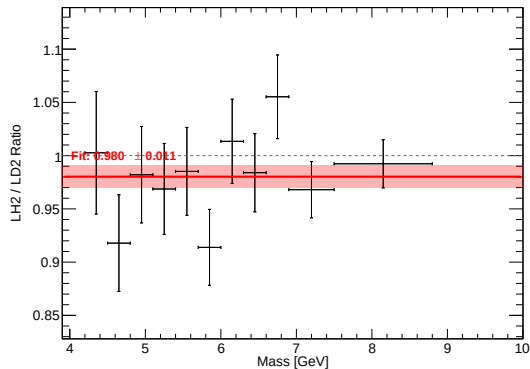


Overlay & Ratio: x_F Bin 15

$0.70 \leq x_F < 0.75$

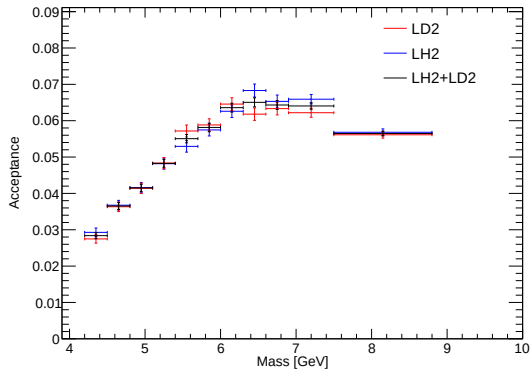


LH2/LD2 $0.70 \leq x_F < 0.75$

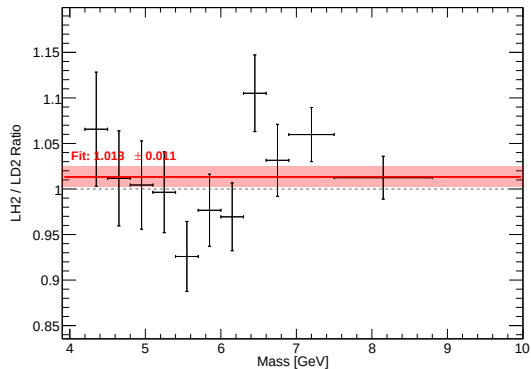


Overlay & Ratio: x_F Bin 16

$0.75 \leq x_F < 0.80$

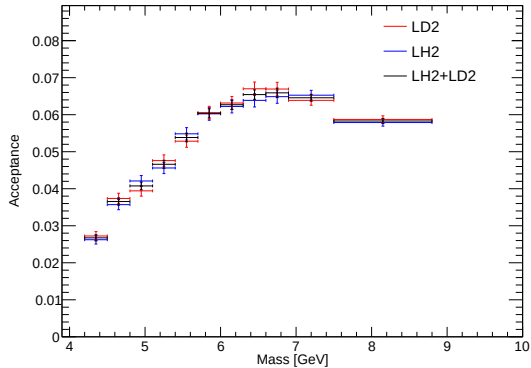


LH2/LD2 $0.75 \leq x_F < 0.80$

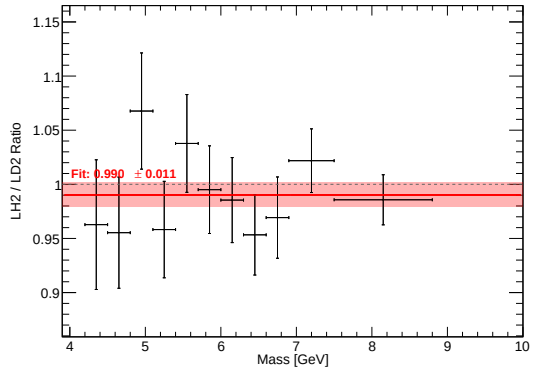


Overlay & Ratio: x_F Bin 17

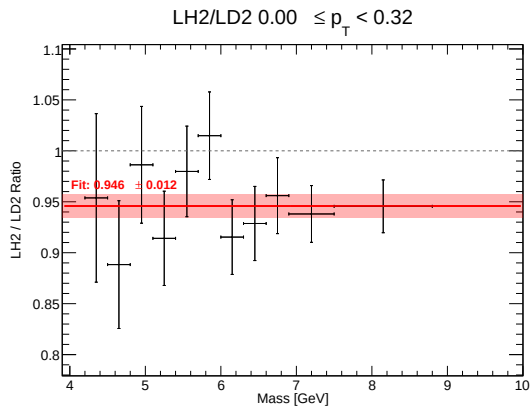
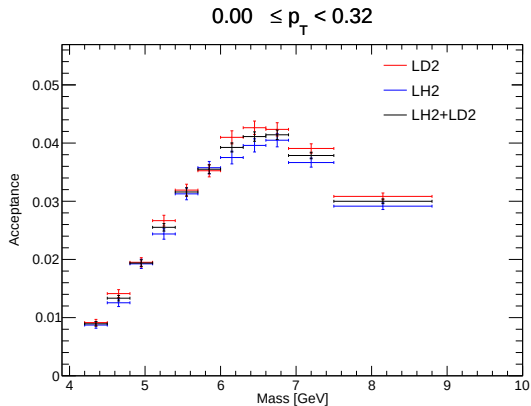
$0.80 \leq x_F < 0.85$



LH2/LD2 $0.80 \leq x_F < 0.85$

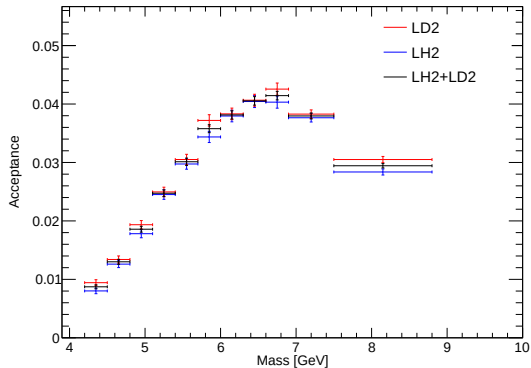


Overlay & Ratio: p_T Bin 0

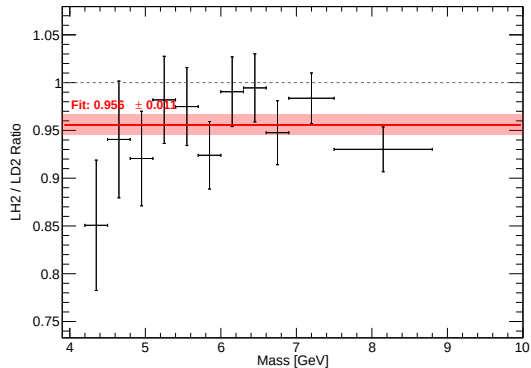


Overlay & Ratio: p_T Bin 1

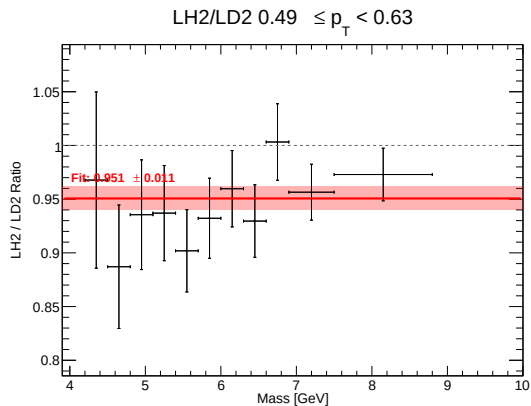
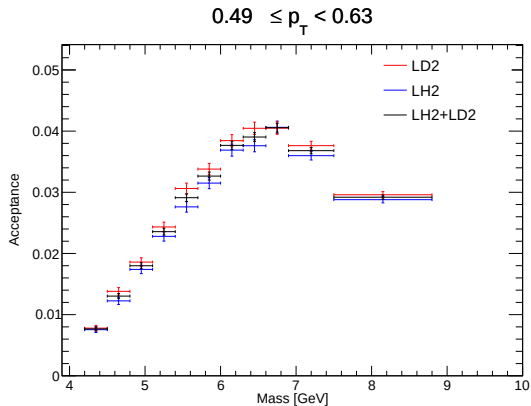
$0.32 \leq p_T < 0.49$



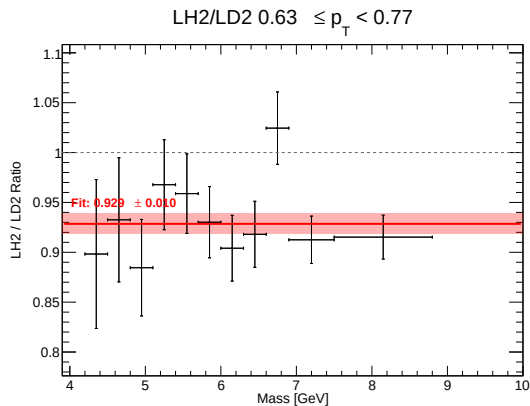
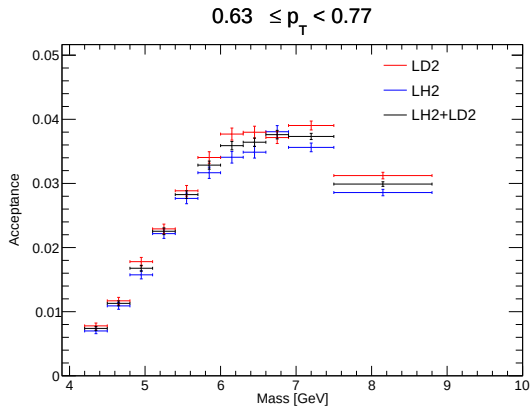
LH2/LD2 $0.32 \leq p_T < 0.49$



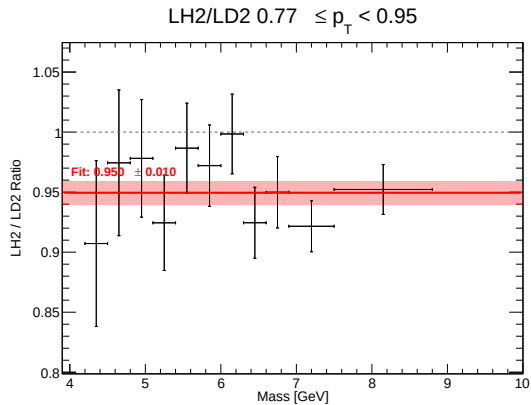
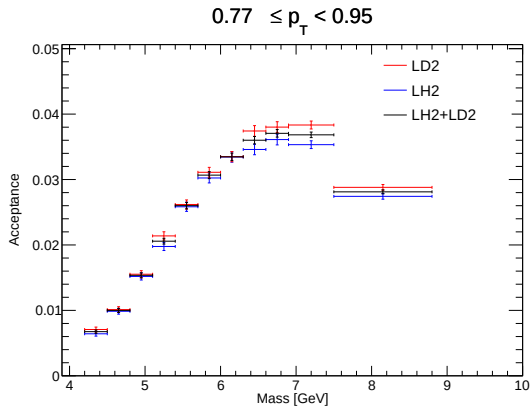
Overlay & Ratio: p_T Bin 2



Overlay & Ratio: p_T Bin 3

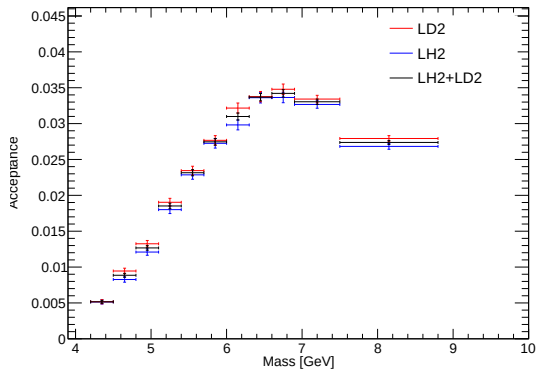


Overlay & Ratio: p_T Bin 4

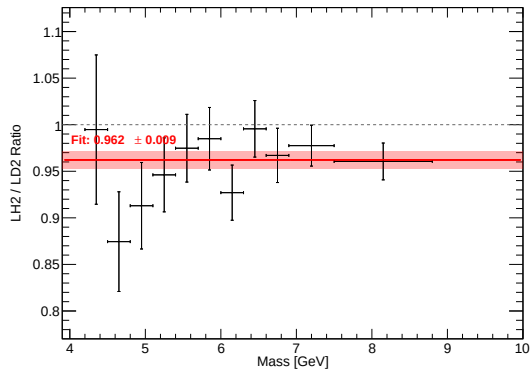


Overlay & Ratio: p_T Bin 5

$0.95 \leq p_T < 1.18$

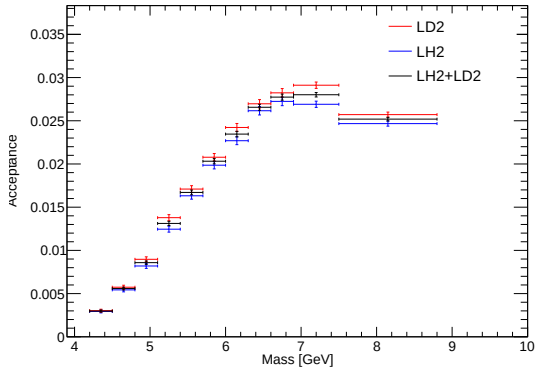


LH2/LD2 $0.95 \leq p_T < 1.18$



Overlay & Ratio: p_T Bin 6

$1.18 \leq p_T < 1.80$



LH2/LD2 $1.18 \leq p_T < 1.80$

